

Hellenic Complex Systems Laboratory

Relation between Diagnostic Accuracy Measures

Technical Report XIV

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Relation between Diagnostic Accuracy Measures

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Short Description of the Demonstration

This Demonstration examines the relation between pairs of diagnostic accuracy measures of diagnostic tests applied to normally distributed nondiseased and diseased populations. This is done for different values of disease prevalence, taking into account the means and standard deviations of the populations. The means and standard deviations are expressed in arbitrary units. The measures considered are the positive predictive value (PPV), the negative predictive value (NPV), the (diagnostic) odds ratio (OR), the likelihood ratio for a positive test result (LR+) and the likelihood test ratio for a negative test result (LR-). The measures can be selected by selecting the respective "plot" and "versus" buttons.

Search Terms

sensitivity, specificity, diagnostic test, diagnostic accuracy, normal distribution, binormal distribution, positive predictive value, negative predictive value, likelihood ratio, odds ratio

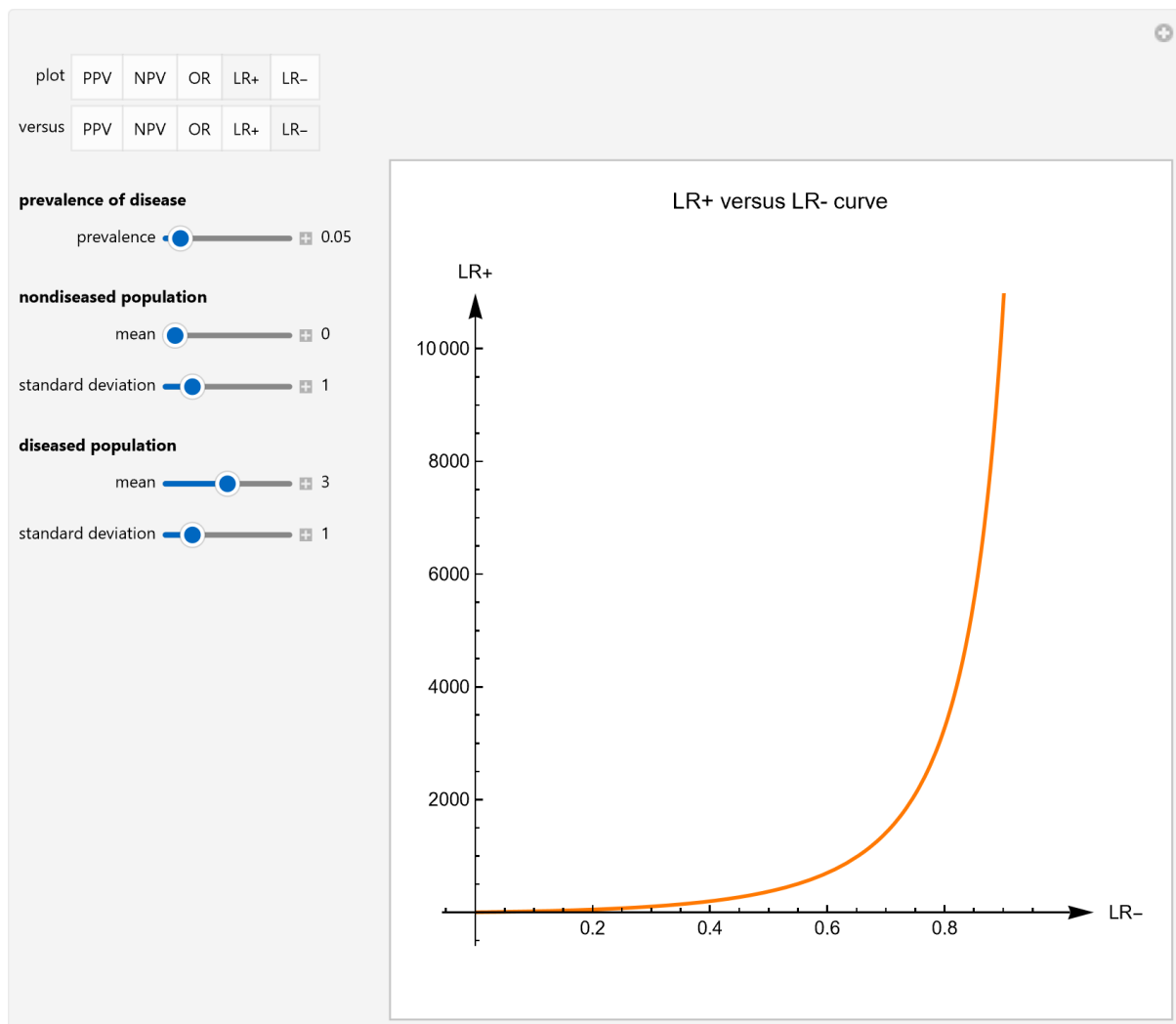


Figure 1: Positive predictive value versus negative predictive value curve of a diagnostic test, with the settings shown at the left.

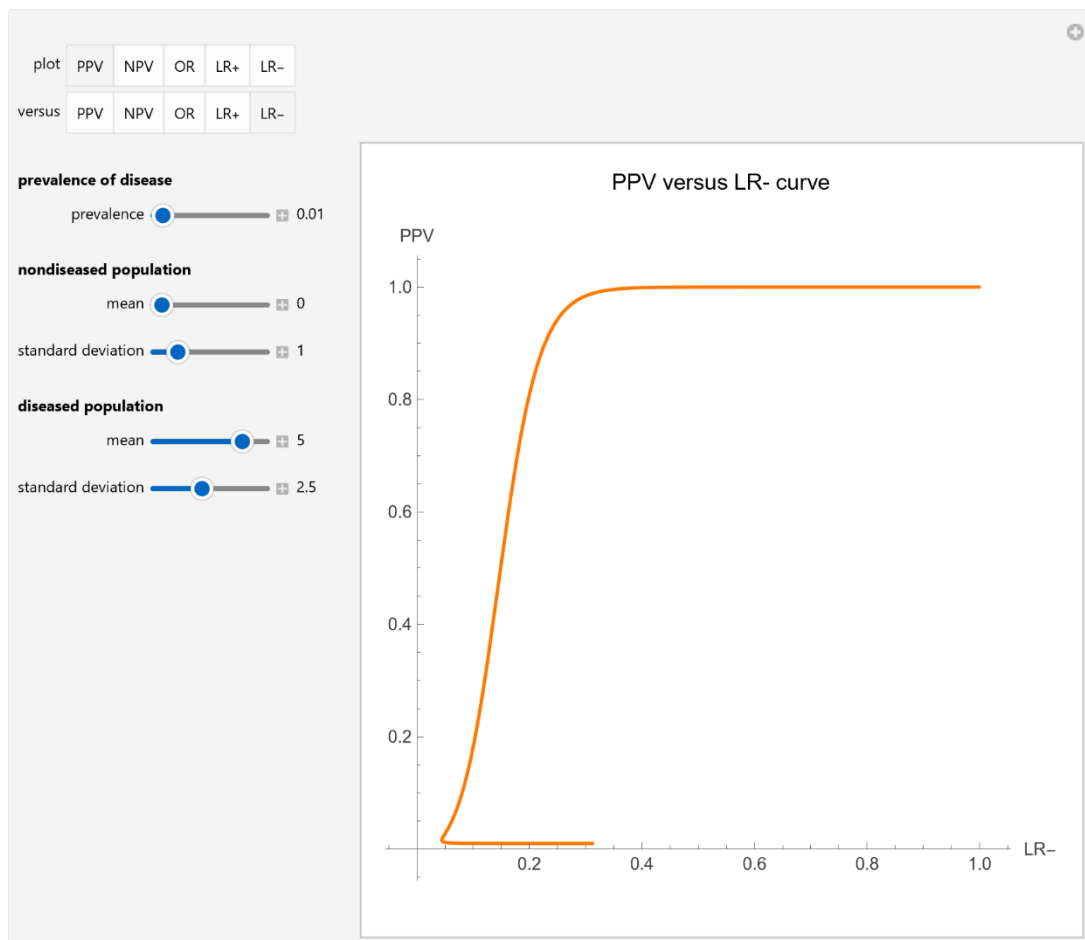


Figure 2: Positive predictive value versus likelihood ratio for a negative test result curve of a diagnostic test, with the settings shown at the left.

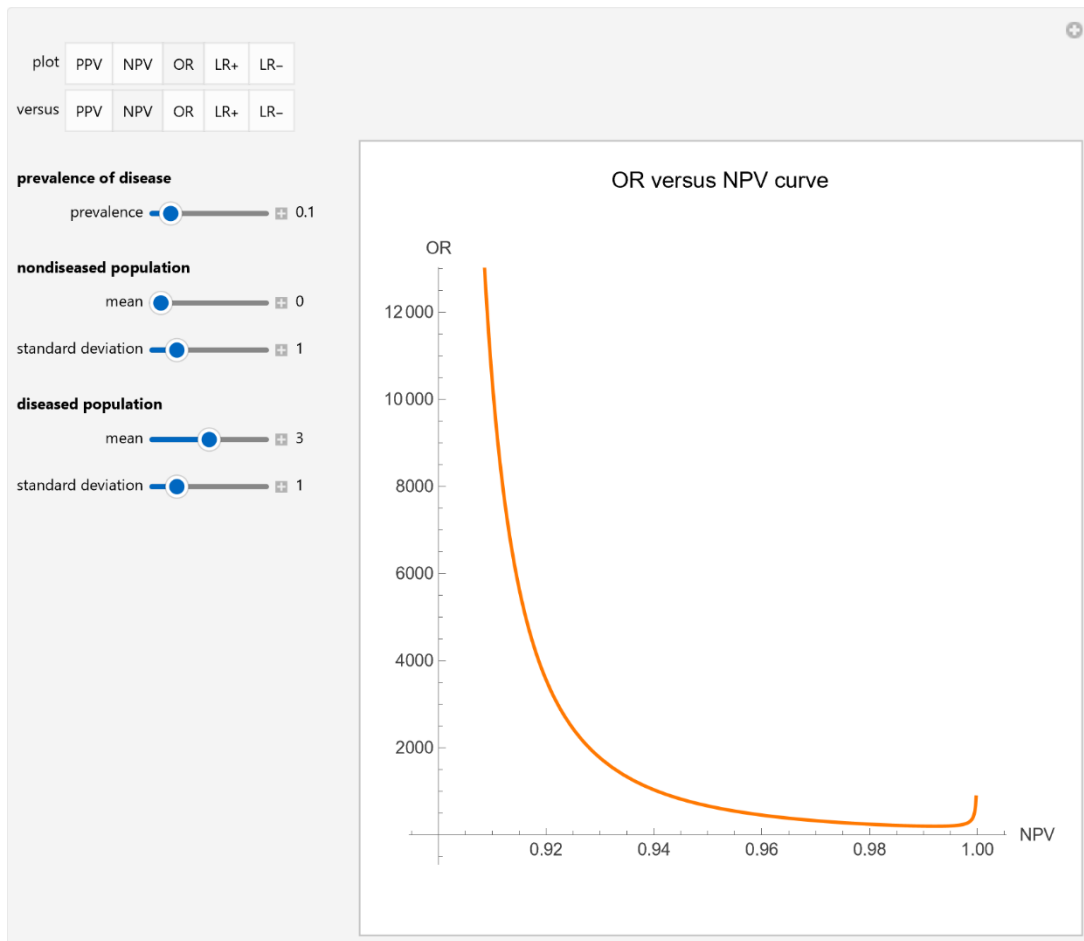


Figure 3: Odds ratio versus negative predictive value curve of a diagnostic test, with the settings shown at the left.

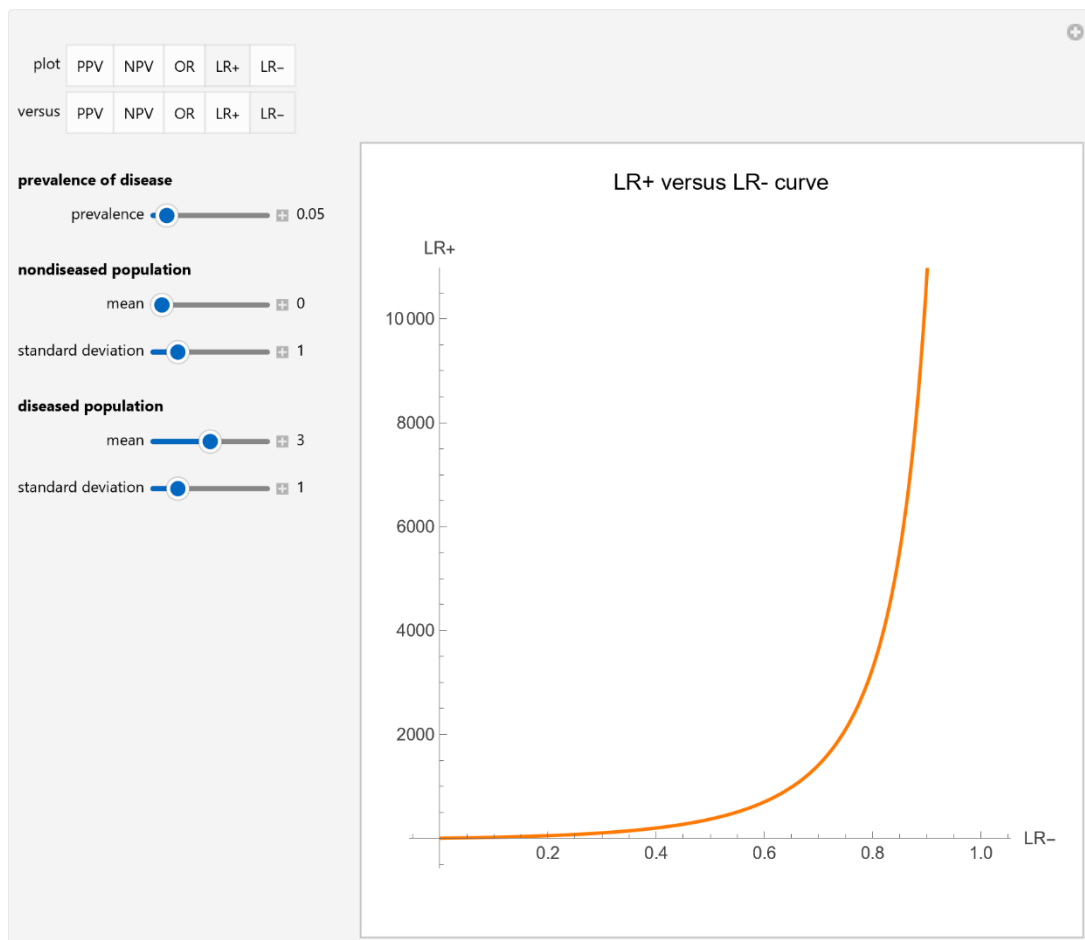


Figure 4: Likelihood ratio for a positive test result versus likelihood ratio for a negative test result curve of a diagnostic test, with the settings shown at the left.

Details

These diagnostic accuracy measures of a diagnostic test applied to a diseased and a nondiseased populations can be calculated as functions of the test sensitivity and specificity. Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. If we denote sensitivity by Se and specificity by Sp the specificity, and by v the prevalence, we have:

$$PPV = \frac{Se v}{Se v + (1 - Sp)(1 - v)}$$
$$NPV = \frac{Sp (1 - v)}{Sp (1 - v) + (1 - v) pr}$$
$$OR = \frac{\frac{Se}{1 - Se}}{\frac{1 - Sp}{Sp}}$$
$$LR+ = \frac{Se}{1 - Sp}$$
$$LR- = \frac{1 - Se}{Sp}$$

Given diseased and a nondiseased populations, the specificity can be defined as a function of sensitivity; therefore, parametric plots can be generated of any pair of the diagnostic accuracy measures.

To the best of our knowledge, with the exception of the pair PPV and NPV [1, 2], the relation between any other pair of the given diagnostic accuracy measures has not been discussed in the literature.

In the thumbnail, the population data describe a bimodal distribution of serum glucose measurements in nondiabetic and diabetic populations [3].

This Demonstration is an extended version of another Demonstration [2], first published in 2015, and is appropriate as an educational tool in medical statistics and as an exploratory research tool in diagnostic accuracy studies.

References

- [1] S.-Y. Shiu and C. Gatsonis. The Predictive Receiver Operating Characteristic Curve for the Joint Assessment of the Positive and Negative Predictive Values. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 366(1874), 2008 pp. 2313–2333. doi:10.1098/rsta.2008.0043.
- [2] Chatzimichail T. *Relation between Positive and Negative Predictive Values*. Ver. Available at: <http://www.hcsl.com/Tools/Demonstrations/RelationOfPositiveAndNegativePredictiveValuesOfDiagnostic.nb>
- [3] T.-O. Lim, R. Bakri, Z. Morad and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care*, 25(12), 2002 pp. 2212–2217. doi:10.2337/diacare.25.12.2212.

Demonstration

DOI

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/RelationOfDiagnosticAccuracyMeasures.nb>

First published in 2018 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/RelationOfDiagnosticAccuracyMeasures/):
<https://demonstrations.wolfram.com/RelationOfDiagnosticAccuracyMeasures/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

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